Angle measurement: estimation errors and measurement noise

2026-10-15 x–z plane, $\theta_x = 0$; continues 1015_wall_angle_readout

1 Estimation errors of the measured quantities

$$\bar{a}_{Tm}[k] - \hat{a}_T[k] = \frac{\delta\tilde{u}_r^2[k] - C_n\,\sigma_n^2}{C_T\,\kappa_T} - \hat{a}_T[k]$$
$$\bar{a}_{Nm}[k] - \hat{a}_N[k] = \frac{\delta\tilde{w}_r^2[k] - C_n\,\sigma_n^2}{C_T\,\kappa_T} - \hat{a}_N[k]$$
$$\theta_{ym}[k] - \hat{\theta}_y[k] = e_{\theta m}[k]$$

$$\mathrm{E}\big[\bar{a}_{Tm} - \hat{a}_T\big] \approx \bar{a}_T - \hat{a}_T$$
$$\mathrm{E}\big[\bar{a}_{Nm} - \hat{a}_N\big] \approx \bar{a}_N - \hat{a}_N$$
$$\mathrm{E}\big[\theta_{ym} - \hat{\theta}_y\big] \approx \frac{\bar{a}_T - \bar{a}_N}{\hat{a}_T - \hat{a}_N}\,e_\theta$$

2 Measurement noise of the measured wall angle

$\xi = C_n\,\sigma_n^2 / C_T\,\kappa_T$; $\delta\tilde{u}_r$, $\delta\tilde{w}_r$ zero-mean Gaussian, nearly uncorrelated (e_θ small); $\rho(\tau)$ of the random component along time

$$\sigma^2\big(\delta\tilde{u}_r\,\delta\tilde{w}_r\big) \approx \sigma^2_{\delta\tilde{u}_r}\,\sigma^2_{\delta\tilde{w}_r} = (C_T\,\kappa_T)^2\,(\bar{a}_T + \xi)(\bar{a}_N + \xi)$$
$$\sigma^2_{e_{\theta m}} = \frac{(\bar{a}_T + \xi)(\bar{a}_N + \xi)}{(\bar{a}_T - \bar{a}_N)^2}$$
$$\rho_{e_{\theta m}}(\tau) = \rho(\tau)^2$$
$$\mathrm{IF} = 1 + 2\sum_{\tau \geq 1} \rho(\tau)^2$$
$$r_\theta = \mathrm{IF} \cdot \frac{(\hat{a}_T + \xi)(\hat{a}_N + \xi)}{(\hat{a}_T - \hat{a}_N)^2}$$